Kubernetes Services

ClusterIP. Exposes a service which is **only accessible from within the cluster**.
NodePort. Exposes a service via a **static port on each node's IP**.
LoadBalancer. Exposes the service via the **cloud provider's load balancer**.
ExternalName. Maps a service to a **predefined externalName field by returning a value for the CNAME** record.

Kubernetes Headless Service - Services that **do not need load balancing and only expose a single IP** can create a 'headless' service by specifying "none" as the clusterIP. **Headless services can be defined with selectors.** in which case endpoint records are created in the API that modify the DNS to return addresses that point to pods that are exposing the service.

NodePorts are **open ports on every cluster node**. Kubernetes will route traffic that comes into a NodePort to the service, even if the service is not running on that node. NodePort is intended as a foundation for other higher-level methods of ingress such as load balancers and are useful in development.

ExternalName services are similar to other Kubernetes services; however, **instead of being accessed via a clusterIP address, it returns a CNAME record with a value** that is defined in the externalName: parameter when creating the service.

Kubernetes services can also be created using the '**kubectl expose**' command, which does not require a YAML file. The same service can be created using the command: **kubectl expose deployment deployment-backend - - port=333- - target-port=4000 - - name=service-backend**

Access a Kubernetes service - There are two ways to discover a Kubernetes service:

DNS (most common): The DNS method is the recommended method of discovering services. To use this method, **a DNS server must first be installed on the cluster**. The DNS server monitors the Kubernetes API, and when a new service is created its name becomes available for easy resolution for requesting applications.
ENV variable: This method **relies on the kubelet adding environment variables for each active service** for every node a pod is running on.

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Kubernetes & Containers

RPS in Application

RPS (Requests Per Second) is a performance metric that measures how many requests an application or system can handle within one second. It is commonly used for measuring the performance and load-bearing capacity of web applications, APIs, or services.

Example Calculation:

If your web server can handle 500 requests every second, the RPS would be 500.

Deploy Ingress Service

An **Ingress** in Kubernetes is a way to expose HTTP and HTTPS services from outside the Kubernetes cluster to services within the cluster.

Example YAML for deploying an Ingress:

yaml

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
name: example-ingress
namespace: default
spec:
rules:
        host: myapp.example.com
        http:
        paths:
        path: /
        pathType: Prefix
        backend:
                service:
                name: myapp-service
                port:
                number: 80
```

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Kubernetes Files

Kubernetes YAML files are used to define the desired state of resources in a Kubernetes cluster. Some of the commonly used files include:

Pod: Defines the containers and their configurations.

Deployment: Describes a deployment strategy for a pod.

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Service: Exposes a set of pods to a network.

Ingress: Configures routing rules to manage external access.

Example of a simple deployment YAML:

yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
name: myapp-deployment
spec:
replicas: 3
selector:
        matchLabels:
        app: myapp
template:
        metadata:
        labels:
        app: myapp
        spec:
        containers:
        name: myapp-container
        image: myapp-image:v1
        ports:
        containerPort: 8080
```

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Terraform EC2 Module

The **Terraform EC2 module** is used to manage EC2 instances in AWS. Example Terraform configuration using a module:

Example:

hcl

```
module "ec2_instance" {
source = "terraform-aws-modules/ec2-instance/aws"
name   = "my-instance"
ami    = "ami-xxxxxxxxxxxxxxxxx"  # Replace with actual AMI ID
instance_type = "t2.micro"
key_name = "my-key"  # Replace with your key
}
```

VDA in Azure

VDA (Virtual Desktop Agent) is a component of **Azure Virtual Desktop (formerly Windows Virtual Desktop)**. It is installed on virtual machines (VMs) that users will connect to and is responsible for managing the virtual desktop experience.

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VDA is required for creating and managing session hosts, and it helps with things like session management, user configuration, and more.

Image Pull Backoff

ImagePullBackOff occurs when Kubernetes fails to pull the image for a container. This could happen due to:

The image does not exist.

Incorrect credentials (for private repositories).

Network issues.

Solution: Verify the image name, repository credentials, and network settings.

Crash Loop Backoff

A **CrashLoopBackOff** occurs when a container repeatedly fails to start and Kubernetes retries it with increasing delays. You can troubleshoot this by checking the container logs:

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```
kubectl logs <pod_name>
```

It can happen due to misconfigurations, missing dependencies, or errors in your container entry point.

Kubernetes Readiness Probe and Liveness Probe

Readiness Probe: Determines if a pod is ready to serve traffic.

Liveness Probe: Checks if a pod is still alive and should be restarted if it fails.

Example:

yaml

```
apiVersion: v1
kind: Pod
metadata:
name: myapp-pod
spec:
containers:
        name: myapp-container
```

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```
image: myapp:v1
readinessProbe:
httpGet:
        path: /healthz
        port: 8080
livenessProbe:
httpGet:
        path: /healthz
        port: 8080
```

Types of Ingress Controller Used in Your Organization - ISTIO

Istio is a service mesh that provides traffic management, security, and observability within microservices. It can also function as an **Ingress controller** to manage incoming traffic.

> **Istio IngressGateway** routes incoming traffic based on defined routing rules and is often used in environments with complex service meshes.

Ingress Flow

The **Ingress flow** involves:

> Traffic from the external world hits the Ingress controller.
>
> The Ingress controller routes the traffic based on configured rules (host, path, etc.).
>
> The traffic is directed to the appropriate backend service within the Kubernetes cluster.

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DaemonSet in Kubernetes

A **DaemonSet** ensures that a specific pod runs on every node or a subset of nodes in the cluster. It is commonly used for running system services like logging, monitoring, etc.

> Example:
>
> yaml
>
> ```
> apiVersion: apps/v1
> kind: DaemonSet
> metadata:
> name: my-daemonset
> spec:
> selector:
>         matchLabels:
>         app: myapp
> template:
> ```

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```
metadata:
labels:
app: myapp
spec:
containers:
name: myapp-container
image: myapp:v1
```

How to Create New Folder in Terraform

You can create a new folder using your terminal/command prompt. For example, on Linux or macOS:

```
mkdir my-terraform-project
```

Ansible Playbook to Install Git

Here's an Ansible playbook that installs Git on Ubuntu:

```
yaml

---
        name: Install Git
hosts: all
become: yes
tasks:
        name: Ensure Git is installed
        apt:
        name: git
        state: present
```

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Self Hosted Agents in Azure

Self-hosted agents are Azure DevOps agents that run on your own infrastructure (VMs, physical servers, etc.). They are typically used for builds, releases, or tests that require specific environments or tools.

To set up:

Go to **Azure DevOps**.

Navigate to **Agent Pools**.

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Install the agent on your server.

Configure the agent to connect to your Azure DevOps account.

Racks in Azure

Azure does not use the concept of "racks" in the traditional data center sense. Instead, it uses **Availability Zones** and **Regions** to achieve fault tolerance, high availability, and redundancy. Availability Zones are isolated locations within a region.

Availability Sets in Azure

Availability Sets ensure that your VMs are distributed across multiple **fault domains** (physical hardware) and **update domains** (planned maintenance). This helps in minimizing downtime during hardware failures or updates.

Fault and Upgrade Domain in Azure

Fault Domain: Represents the physical hardware that could fail (e.g., power, networking). VMs in the same fault domain could be affected by the same failure.

Upgrade Domain: Refers to the grouping of VMs to ensure that updates are applied to a small batch of VMs at a time to prevent complete downtime.

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Upgrade Plugin in Terraform

To upgrade all your provider plugins, run:

```
terraform init -upgrade
```

Page Blobs in Azure

Page Blobs are used in Azure to store virtual machine disk images. They allow for **random read/write operations** and are optimized for virtual machine use cases.

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Virtual HD in Azure

Virtual Hard Disks (VHDs) are used in Azure to store the operating system and data for virtual machines. They can be stored as blobs in Azure Storage.

Shell Script to Find All Prime Numbers 1 to 10

Here's a simple script to find prime numbers between 1 and 10:

```
for num in {1..10}; do
is_prime=1
for ((i=2; i<num; i++)); do
        if ((num % i == 0)); then
        is_prime=0
        break
        fi
done
if ((is_prime == 1)); then
        echo $num
fi
done
```

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What is a Virtual Private Cloud (VPC)?

A **Virtual Private Cloud (VPC)** is a logically isolated network within a cloud provider's environment. It allows you to define your own IP address range, subnets, route tables, and network gateways. It can be used to run resources like EC2 instances, RDS, and Lambda functions with full control over the network architecture.

What is an Elastic Load Balancer (ELB)?

Elastic Load Balancer (ELB) distributes incoming traffic across multiple targets such as EC2 instances, containers, and IP addresses. Types include **Classic Load Balancer**, **Application Load Balancer (ALB)**, and **Network Load Balancer (NLB)**, catering to different use cases like Layer 7 (ALB) or Layer 4 (NLB) routing.

What is the difference between EC2 and Lambda?

EC2: Scalable virtual machines with full control over the operating system, ideal for custom applications and configurations.

Lambda: Serverless compute service where you run code in response to events without managing servers. You pay only for the execution time.

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What is CloudFormation?

CloudFormation is AWS's Infrastructure as Code (IaC) service, which allows you to provision and manage AWS resources using templates (written in YAML or JSON), enabling automated and repeatable infrastructure deployments.

What is the difference between CloudFormation and Terraform?

CloudFormation is AWS-specific and uses templates to define infrastructure.

Terraform is multi-cloud, supports various providers (AWS, Azure, GCP), and uses **HashiCorp Configuration Language (HCL)** for resource definitions.

What is an S3 bucket?

Amazon S3 (Simple Storage Service) is object storage designed to store and retrieve any amount of data at any time. S3 buckets are the containers for storing objects (files) and can be publicly or privately accessed.

What is a VPC Subnet?

A **VPC Subnet** is a segment of a VPC's IP address range where you can place your AWS resources, such as EC2 instances. Subnets can be private or public depending on their routing configuration and whether they can access the internet.

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How does Elastic IP work?

Elastic IP is a static IPv4 address associated with your AWS account, allowing you to reassign the address to any EC2 instance in the same region, offering failover capabilities for highly available applications.

What is AWS IAM?

AWS Identity and Access Management (IAM) allows you to securely control access to AWS resources. It lets you create users, groups, and roles with specific permissions to access AWS services and resources.

What are IAM Roles?

IAM Roles are AWS identities with specific permissions that can be assumed by AWS resources (e.g., EC2 instances, Lambda functions) or users to perform actions on resources.

CI/CD and Automation

What is CI/CD?

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CI (Continuous Integration) is the practice of automatically integrating code into a shared repository multiple times per day.

CD (Continuous Delivery) automates the deployment of integrated code to production or staging environments, reducing manual intervention.

What is Jenkins and its use in DevOps?

Jenkins is an open-source automation server used to set up continuous integration/continuous delivery (CI/CD) pipelines. It automates tasks like building, testing, and deploying applications.

What is Docker and how does it relate to CI/CD?

Docker is a platform that uses containerization to deploy and manage applications in isolated environments. In CI/CD, Docker allows building, testing, and deploying applications in consistent environments.

What is Kubernetes?

Kubernetes is an open-source platform for automating deployment, scaling, and operations of containerized applications. It helps manage clusters of containers, ensuring high availability, scaling, and load balancing.

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What is a Kubernetes Pod?

A **Pod** is the smallest deployable unit in Kubernetes, consisting of one or more containers that share the same network namespace and storage.

What is a Jenkins Pipeline?

A **Jenkins Pipeline** is a suite of plugins that allows users to define a complete build and deployment pipeline as code, either in a **Declarative** or **Scripted** format.

What is Docker Compose?

Docker Compose is a tool for defining and running multi-container Docker applications. It allows you to specify multiple containers and their configurations using a YAML file.

What are the different types of Kubernetes Services?

ClusterIP: Default service type, exposes the service on a cluster-internal IP.

NodePort: Exposes the service on a static port on each node's IP.

LoadBalancer: Exposes the service externally using a cloud provider’s load balancer.

ExternalName: Maps the service to an external DNS name.

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What is a Kubernetes Deployment?

A **Deployment** provides declarative updates to Pods and ReplicaSets. It allows you to manage stateless applications, scaling, and rollout of updates.

What is Helm in Kubernetes?

Helm is a package manager for Kubernetes that allows users to define, install, and manage Kubernetes applications via **charts**, which are pre-configured sets of Kubernetes resources.

What is a Docker Image?

A **Docker Image** is a read-only template used to create containers. It includes everything needed to run an application, such as the code, libraries, dependencies, and the runtime environment.

Monitoring & Logging

What is Prometheus?

Prometheus is an open-source monitoring system designed for reliability and scalability. It collects time-series data (metrics) from applications, infrastructure, and services.

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What is Grafana?

Grafana is an open-source tool used for visualizing time-series data, often paired with Prometheus to display metrics through customizable dashboards.

What is the ELK Stack?

The **ELK Stack** consists of:

Elasticsearch: Search and analytics engine.

Logstash: A server-side pipeline that collects, parses, and stores logs.

Kibana: Data visualization platform for analyzing and visualizing data stored in Elasticsearch.

What is Fluentd?

Fluentd is an open-source log collector and aggregator used to unify data collection and consumption. It helps in transforming and routing log data to various backends (like Elasticsearch, Kafka, etc.).

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What is AWS CloudWatch?

AWS CloudWatch is a monitoring and observability service for AWS resources. It collects metrics, logs, and events, and helps in setting alarms for various resources to trigger automated actions.

Security in DevOps

What is DevSecOps?

DevSecOps integrates security practices into the DevOps pipeline, ensuring security is a shared responsibility throughout the development lifecycle. It includes practices such as vulnerability scanning, threat modeling, and code analysis.

What is Role-Based Access Control (RBAC)?

RBAC is a security mechanism that restricts system access based on users' roles within an organization. It ensures users only have access to the resources required for their tasks.

What is SSL/TLS?

SSL (Secure Sockets Layer) and **TLS (Transport Layer Security)** are cryptographic protocols used to secure communication over networks, primarily for web traffic (HTTPS).

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What is Two-Factor Authentication (2FA)?

Two-Factor Authentication (2FA) requires two different forms of identification for logging in—something you know (password) and something you have (e.g., a phone or hardware token).

What is a Security Token Service (STS)?

STS is an AWS service that provides temporary security credentials, allowing users or services to authenticate with AWS resources for a limited time.

Infrastructure Automation

What is Ansible?

Ansible is an open-source automation tool used for configuration management, application deployment, and task automation. It uses **YAML** for writing playbooks.

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What is Puppet?

Puppet is a configuration management tool that automates infrastructure management tasks by defining system configurations through code, allowing for consistent and repeatable setups.

What is Terraform?

Terraform is an open-source IaC (Infrastructure as Code) tool for defining and provisioning infrastructure using configuration files. It supports multiple cloud providers like AWS, Azure, GCP, and others.

What is Chef?

Chef is an automation tool for managing infrastructure through code. It uses **recipes** written in Ruby to define configurations for servers and applications.

Networking & Security

What is a Security Group in AWS?

Security Groups in AWS are virtual firewalls that control inbound and outbound traffic to EC2 instances. They work at the instance level and allow you to specify rules based on IP address, port, and protocol.

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What is a Network ACL (NACL)?

Network Access Control Lists (NACLs) are stateless firewalls that operate at the subnet level in a VPC. They provide an additional layer of security by controlling inbound and outbound traffic for subnets.

What is the difference between a Security Group and NACL?

Security Groups are stateful and control traffic to and from instances, while **NACLs** are stateless and control traffic to and from subnets. Security groups allow more granular control at the instance level, whereas NACLs operate at the subnet level.

What is AWS Direct Connect?

AWS Direct Connect is a service that establishes a dedicated, high-bandwidth, low-latency network connection between your on-premises data center and AWS, improving performance and security.

What is a VPN Gateway in AWS?

VPN Gateway is a VPN concentrator on the AWS side of a VPN connection, used to establish secure connections between your VPC and on-premises network or other VPCs.

What is AWS Shield?

AWS Shield is a managed Distributed Denial of Service (DDoS) protection service. It provides protection against common attacks like SYN/ACK floods and offers advanced protections (Shield Advanced) for larger-scale attacks.

What is AWS WAF?

AWS Web Application Firewall (WAF) helps protect web applications from common web exploits like SQL injection, cross-site scripting, and other vulnerabilities by allowing you to define custom rules to block harmful traffic.

What is a Bastion Host?

A **Bastion Host** is a server that acts as a gateway between a secure network and an untrusted network, usually for administrative access to EC2 instances in private subnets.

What is a Load Balancer?

A **Load Balancer** distributes network traffic across multiple servers to ensure no single server is overwhelmed, improving availability and reliability.

Databases & Data Management

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What is Amazon RDS?

Amazon RDS (Relational Database Service) is a managed relational database service that supports databases like MySQL, PostgreSQL, MariaDB, Oracle, and SQL Server. It automates backups, patch management, and scaling.

What is Amazon DynamoDB?

Amazon DynamoDB is a fully managed NoSQL database that provides fast and predictable performance with scalability. It is designed for applications that require low-latency data access.

What is Amazon Redshift?

Amazon Redshift is a fully managed data warehouse service that allows you to run complex queries on large datasets. It is optimized for OLAP (Online Analytical Processing) workloads.

What is Amazon Aurora?

Amazon Aurora is a high-performance, fully managed relational database engine compatible with MySQL and PostgreSQL. It provides better performance and availability than traditional MySQL and PostgreSQL.

What is Amazon ElastiCache?

Amazon ElastiCache is a fully managed caching service that supports Redis and Memcached. It helps improve the performance of web applications by storing frequently accessed data in memory.

What is Amazon SQS?

Amazon SQS (Simple Queue Service) is a fully managed message queuing service that enables you to decouple and scale microservices, distributed systems, and serverless applications.

What is Amazon SNS?

Amazon SNS (Simple Notification Service) is a fully managed messaging service that allows you to send messages or notifications to a large number of recipients via SMS, email, or push notifications.

What is AWS DMS?

AWS Database Migration Service (DMS) helps migrate databases to AWS quickly and securely. It supports migrations from on-premises databases to RDS, Aurora, and other database services.

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What is AWS Glue?

AWS Glue is a fully managed ETL (Extract, Transform, Load) service that prepares and loads data for analytics. It automatically discovers and catalogs data stored in AWS services like S3 and RDS.

What is Amazon Kinesis?

Amazon Kinesis is a platform for real-time processing of streaming data at scale. It provides services like Kinesis Streams, Kinesis Firehose, and Kinesis Analytics to collect, process, and analyze data streams.

What is AWS Athena?

AWS Athena is an interactive query service that allows you to analyze data stored in Amazon S3 using SQL. It is serverless and requires no infrastructure management.

Cost Management & Optimization

What is AWS Cost Explorer?

AWS Cost Explorer is a tool that enables you to visualize and analyze your AWS spending patterns. It helps identify cost trends, set budgets, and forecast future expenses.

What is AWS Trusted Advisor?

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AWS Trusted Advisor is an online resource to help optimize your AWS infrastructure. It provides real-time recommendations based on best practices in areas like cost optimization, security, and fault tolerance.

What is AWS Budgets?

AWS Budgets allows you to set custom cost and usage budgets and receive alerts when your usage or spending exceeds your thresholds. This helps track and control AWS costs.

What is Reserved Instances (RIs)?

Reserved Instances (RIs) are a pricing model in AWS that allows you to reserve capacity for EC2 instances for a one- or three-year term, offering significant savings compared to on-demand pricing.

What is Spot Instances?

Spot Instances allow you to bid for unused EC2 capacity at a lower price. These instances can be interrupted by AWS with a two-minute warning, making them suitable for fault-tolerant and flexible applications.

What is AWS Savings Plans?

AWS Savings Plans offer flexible pricing models that allow you to save up to 72% on compute usage in exchange for committing to a consistent usage level (e.g., compute usage for 1 or 3 years).

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Application & Deployment Management

What is AWS Elastic Beanstalk?

AWS Elastic Beanstalk is a Platform as a Service (PaaS) offering for deploying and managing applications. It abstracts the infrastructure layer and handles deployment, scaling, and monitoring.

What is Amazon Lightsail?

Amazon Lightsail is an easy-to-use cloud platform designed for developers, offering simple virtual private servers, storage, and networking. It’s ideal for users looking for an affordable alternative to EC2.

What is AWS CodePipeline?

AWS CodePipeline is a fully managed continuous integration and continuous delivery (CI/CD) service that automates the build, test, and deploy phases of your application release process.

What is AWS CodeCommit?

AWS CodeCommit is a fully managed source control service that allows you to host private Git repositories. It provides version control for collaborative software development.

What is AWS CodeDeploy?

AWS CodeDeploy is a service that automates application deployment to various compute services, including EC2 instances, Lambda functions, and on-premises servers.

What is AWS CodeBuild?

AWS CodeBuild is a fully managed build service that compiles your source code, runs tests, and produces software packages ready for deployment.

What is AWS SAM (Serverless Application Model)?

AWS SAM is an open-source framework that simplifies the development of serverless applications on AWS. It provides shorthand syntax for defining resources like AWS Lambda functions and API Gateway.

What is Amazon API Gateway?

Amazon API Gateway is a fully managed service that allows you to create, publish, and monitor RESTful APIs for your applications, often used for serverless architectures with Lambda functions.

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What is AWS AppSync?

AWS AppSync is a fully managed GraphQL service that simplifies application development by enabling real-time data synchronization, offline access, and backend data aggregation.

Serverless Computing

What is Serverless Computing?

Serverless Computing is a cloud computing model where cloud providers automatically manage the infrastructure required to run your applications. AWS services like Lambda, API Gateway, and DynamoDB enable serverless architectures, where you focus on writing code and don't worry about provisioning servers.

What is AWS Lambda?

AWS Lambda is a serverless compute service that allows you to run code in response to events without provisioning or managing servers. You only pay for the execution time.

What are AWS Step Functions?

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AWS Step Functions is a serverless orchestration service that allows you to design and coordinate the components of distributed applications and microservices. It integrates with AWS Lambda and other services for building workflows.

What is AWS Fargate?

AWS Fargate is a serverless compute engine for containers. It allows you to run containers without managing the underlying EC2 instances, making it easier to scale and manage containerized applications.

What is a VPC in AWS?

VPC (Virtual Private Cloud) allows you to create a logically isolated section of AWS where you can launch AWS resources in a virtual network. You can control IP address ranges, subnets, route tables, and network gateways.

What are the different types of VPC Peering?

VPC Peering enables communication between two VPCs. There are two types of VPC peering:

- **One-to-One Peering**: Between two VPCs within the same region or across regions.
- **Transitive Peering**: Peering between three or more VPCs through a central hub VPC.

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What is the difference between Public and Private Subnets in AWS?

Public Subnet has a route to the internet via an Internet Gateway, allowing resources like EC2 instances to access the internet.

Private Subnet does not have direct access to the internet, but it can use a NAT Gateway or NAT instance to initiate outbound connections.

What is an Internet Gateway?

Internet Gateway (IGW) is a VPC component that allows communication between instances in your VPC and the internet. It's necessary for instances in a public subnet to access the internet.

What is a NAT Gateway?

NAT Gateway allows instances in a private subnet to initiate outbound internet traffic (e.g., to download patches) while preventing inbound internet traffic from reaching those instances.

What is AWS Transit Gateway?

AWS Transit Gateway is a hub-and-spoke model for interconnecting VPCs, on-premises networks, and remote offices. It simplifies network management and scaling.

Storage & Content Delivery

What is Amazon S3?

Amazon S3 (Simple Storage Service) is an object storage service that provides scalable and durable storage for large amounts of data. It allows you to store any type of data and access it from anywhere via the web.

What is S3 Versioning?

S3 Versioning enables you to keep multiple versions of an object in a bucket. It helps protect data from accidental deletion or overwriting.

What is Amazon Glacier?

Amazon Glacier is a low-cost archival storage service for data that is infrequently accessed. It provides durable storage but requires longer retrieval times compared to S3.

What is AWS EBS?

AWS EBS (Elastic Block Store) provides block-level storage that can be attached to EC2 instances. It is suitable for use cases requiring a file system, database storage, or access to raw data.

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What is EBS Snapshots?

EBS Snapshots are point-in-time backups of EBS volumes. They are stored in S3 and allow you to create a copy of your volume for backup or disaster recovery.

What is AWS Storage Gateway?

AWS Storage Gateway is a hybrid cloud storage service that connects on-premises environments to AWS storage services. It supports file, volume, and tape gateway configurations.

What is Amazon FSx?

Amazon FSx provides fully managed file systems. FSx for Windows File Server is for Windows-based applications, and FSx for Lustre is optimized for high-performance workloads like big data and ML.

What is Amazon CloudFront?

Amazon CloudFront is a content delivery network (CDN) that accelerates the delivery of your websites, videos, APIs, and other content to users worldwide.

What is AWS Snowball?

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AWS Snowball is a data transport solution that allows you to transfer large amounts of data into and out of AWS. It is available in two options: Snowball Edge (with compute capabilities) and Snowball (for large data transfers).

Automation & Orchestration

What is AWS CloudFormation?

AWS CloudFormation is an infrastructure as code (IaC) service that allows you to define and provision AWS infrastructure resources in a declarative manner using templates (JSON or YAML).

What is AWS CloudTrail?

AWS CloudTrail is a service that records and logs all API calls made within an AWS account, allowing you to track user activity and changes to your AWS resources for security and auditing.

What is AWS Config?

AWS Config is a service that provides a detailed inventory of your AWS resources and their configuration history. It allows you to assess, audit, and evaluate the configurations of your AWS resources.

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What is AWS OpsWorks?

AWS OpsWorks is a configuration management service that uses Chef and Puppet to automate infrastructure management tasks, allowing you to manage servers and applications.

What is AWS Elastic Load Balancing (ELB)?

Elastic Load Balancing (ELB) automatically distributes incoming traffic across multiple EC2 instances in one or more Availability Zones, ensuring fault tolerance and scalability.

What is AWS Auto Scaling?

AWS Auto Scaling automatically adjusts the number of EC2 instances in your environment based on traffic demands. It ensures that you have the right number of instances running to handle the load.

Developer Tools & Services

What is AWS CodeStar?

AWS CodeStar is a fully integrated development environment (IDE) that enables you to quickly develop, build, and deploy applications on AWS. It integrates with other AWS developer tools like CodePipeline and CodeCommit.

What is AWS Cloud9?

AWS Cloud9 is a cloud-based integrated development environment (IDE) that provides a rich code-editing experience and enables collaboration with team members.

What is Amazon Elastic Container Service (ECS)?

Amazon ECS is a fully managed container orchestration service that supports Docker containers. It allows you to run applications in a scalable and secure manner without managing the underlying infrastructure.

What is AWS ECR?

AWS ECR (Elastic Container Registry) is a fully managed Docker container registry service where you can store, manage, and deploy Docker images for applications.

What is AWS Lambda@Edge?

AWS Lambda@Edge allows you to run Lambda functions at AWS CloudFront edge locations. This enables you to execute functions closer to the users, reducing latency.

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Big Data & Analytics

What is Amazon EMR?

Amazon EMR (Elastic MapReduce) is a managed cluster platform for running big data frameworks like Apache Hadoop, Apache Spark, and others. It simplifies processing and analyzing large amounts of data.

What is AWS Glue Data Catalog?

AWS Glue Data Catalog is a centralized metadata repository that stores information about the data in your data lake or data warehouse, enabling seamless integration with other AWS analytics services.

What is Amazon QuickSight?

Amazon QuickSight is a scalable, business intelligence (BI) service that provides data visualization, reporting, and analytics capabilities for your AWS data.

What is AWS Lake Formation?

AWS Lake Formation is a service that simplifies the process of creating and managing data lakes, enabling secure and governed data access for analytics and machine learning workloads.

What is AWS X-Ray?

AWS X-Ray is a distributed tracing service that helps developers analyze and debug the performance of their applications, identifying bottlenecks and optimizing the user experience.

What is AWS Athena?

AWS Athena is an interactive query service that makes it easy to analyze data in Amazon S3 using SQL. It is serverless and allows you to start querying without setting up or managing infrastructure.

What is Amazon Kinesis Firehose?

Amazon Kinesis Firehose is a fully managed service that can capture, transform, and load real-time data streams into data stores like Amazon S3, Redshift, or Elasticsearch.

What is Amazon Kinesis Data Streams?

Amazon Kinesis Data Streams allows you to collect and process real-time data streams at massive scale. It can be used for analytics, monitoring, and log aggregation.

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Machine Learning

What is Amazon SageMaker?

Amazon SageMaker is a fully managed service that enables developers and data scientists to build, train, and deploy machine learning models quickly.

What is Amazon Comprehend?

Amazon Comprehend is a natural language processing (NLP) service that uses machine learning to analyze and understand text, including sentiment analysis and entity recognition.

What is Amazon Rekognition?

Amazon Rekognition is a machine learning service that allows you to analyze and recognize objects, scenes, and faces in images and videos.

What is Amazon Polly?

Amazon Polly is a text-to-speech service that converts written text into lifelike speech using deep learning.

What is Amazon Lex?

Amazon Lex is a service for building conversational interfaces into applications using voice and text. It powers Alexa and helps in creating chatbots.

What is Amazon Translate?

Amazon Translate is a neural machine translation service that delivers high-quality translations for text in real time.

Migration & Hybrid Cloud

What is AWS Migration Hub?

AWS Migration Hub provides a single location to track the progress of application migrations across multiple AWS and partner tools. It provides visibility and tracking for migrations.

What is AWS Database Migration Service (DMS)?

AWS DMS helps you migrate databases to AWS with minimal downtime. It supports homogenous and heterogeneous migrations between various database engines.

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What is AWS Server Migration Service (SMS)?

AWS SMS automates the migration of on-premises servers to AWS. It supports incremental replication for fast and efficient server migration.

What is AWS Snowball Edge?

AWS Snowball Edge is a physical device with compute and storage capabilities, used to transfer large amounts of data to AWS or run edge computing workloads in remote locations.

Pricing, Billing & Cost Management

What is AWS Cost Explorer?

AWS Cost Explorer is a tool that allows you to visualize, analyze, and understand your AWS spending patterns. You can track usage, cost, and forecast future expenses.

What is AWS Trusted Advisor?

AWS Trusted Advisor is an online resource to help you reduce cost, increase performance, and improve security by providing best practice recommendations for your AWS resources.

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What is the AWS Free Tier?

AWS Free Tier provides limited free usage of AWS services for 12 months after you create an AWS account. It includes services like EC2, S3, and Lambda with usage limits.

What is AWS Budgets?

AWS Budgets allows you to set custom cost and usage budgets for your AWS resources and receive alerts when your usage or costs exceed defined thresholds.

What is the AWS Pricing Calculator?

AWS Pricing Calculator is an online tool to help you estimate the cost of your AWS architecture by configuring resources and selecting services.

What are AWS Reserved Instances?

AWS Reserved Instances allow you to commit to a one- or three-year term for certain AWS services (e.g., EC2, RDS) to receive significant discounts compared to on-demand pricing.

General Concepts

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What is an IAM role?

An **IAM role** is an identity within AWS with specific permissions that can be assumed by an AWS service or an authenticated user to perform actions on AWS resources.

What is an IAM policy?

IAM policy is a document that defines permissions to access AWS resources. Policies are attached to IAM users, groups, or roles.

What is an AWS Availability Zone?

An **Availability Zone (AZ)** is a physically isolated data center within a region, designed to be independent and fault-tolerant to provide high availability.

What is an AWS Region?

An **AWS Region** is a geographical location where AWS data centers are located. Each region consists of multiple Availability Zones (AZs).

What is a CloudFormation Template?

A **CloudFormation template** defines the resources and their configurations that make up an AWS infrastructure stack. It is written in JSON or YAML format.

What is Amazon EC2 Auto Scaling?

EC2 Auto Scaling automatically adjusts the number of EC2 instances in your fleet based on demand. It ensures that you have the right amount of resources to handle traffic spikes.

What is AWS Direct Connect?

AWS Direct Connect is a service that allows you to establish a dedicated, private network connection from your on-premises data center to AWS. It reduces latency, increases bandwidth, and can lower costs compared to internet-based connections.

What is Security Group in AWS?

Security Groups act as virtual firewalls for your EC2 instances, controlling inbound and outbound traffic based on defined rules (e.g., IP address ranges, protocols, and ports).

What is AWS WAF (Web Application Firewall)?

AWS WAF is a managed web application firewall that helps protect your web applications from common web exploits, such as SQL injection or cross-site scripting (XSS) attacks.

What is AWS Shield?

AWS Shield is a managed DDoS protection service that safeguards AWS applications from distributed denial-of-service (DDoS) attacks, providing enhanced protection against these threats.

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Storage & Content Delivery (continued)

What is Amazon EFS (Elastic File System)?

Amazon EFS is a fully managed file storage service that allows you to create and configure scalable, elastic file systems for use with Amazon EC2 instances, accessible from multiple instances simultaneously.

What is AWS Backup?

AWS Backup is a fully managed backup service that allows you to centralize and automate data backup across AWS services like EBS, RDS, DynamoDB, and more.

What is S3 Transfer Acceleration?

S3 Transfer Acceleration is a service that speeds up the upload and download of data to and from S3 buckets by routing traffic through CloudFront edge locations.

Developer Tools & Services (continued)

What is AWS CodePipeline?

AWS CodePipeline is a continuous integration and delivery (CI/CD) service that automates the build, test, and deploy phases of your application's lifecycle, enabling faster and more reliable software releases.

What is AWS CodeCommit?

AWS CodeCommit is a fully managed source control service that supports Git repositories, allowing developers to store and version control their source code in the cloud.

What is AWS CodeBuild?

AWS CodeBuild is a fully managed build service that compiles source code, runs tests, and produces artifacts ready for deployment to AWS services.

Machine Learning (continued)

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What is Amazon Forecast?

Amazon Forecast is a fully managed service that uses machine learning to generate highly accurate time-series forecasts based on historical data, which can be used for demand planning, inventory management, and more.

What is Amazon Textract?

Amazon Textract is a machine learning service that automatically extracts text, forms, and tables from scanned documents. It helps you automate document processing workflows.

What is Amazon Personalize?

Amazon Personalize is a machine learning service that enables developers to build real-time personalized recommendation systems for their users.

What is AWS Deep Learning AMI?

AWS Deep Learning AMI provides a pre-configured environment with popular deep learning frameworks like TensorFlow, PyTorch, and Apache MXNet. It allows you to quickly deploy and scale deep learning models in the cloud.

Big Data & Analytics (continued)

What is Amazon Redshift?

Amazon Redshift is a fully managed, petabyte-scale data warehouse service designed for online analytic processing (OLAP). It allows fast querying and analysis of large datasets.

What is Amazon Kinesis?

Amazon Kinesis is a platform for real-time data streaming. It includes several services like Kinesis Data Streams, Kinesis Data Firehose, and Kinesis Data Analytics for handling real-time data collection, processing, and analysis.

What is AWS Data Pipeline?

AWS Data Pipeline is a web service that allows you to automate the movement and transformation of data between different AWS compute and storage services, such as S3, Redshift, and RDS.

Migration & Hybrid Cloud (continued)

What is AWS Database Migration Service (DMS)?

AWS DMS helps you migrate databases to AWS with minimal downtime. It supports both homogenous (same engine) and heterogeneous (different engine) database migrations and continuous replication.

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What is AWS Application Discovery Service?

AWS Application Discovery Service helps enterprises plan their migration to AWS by discovering and understanding their on-premises workloads. It collects information about applications, servers, and dependencies.

What is AWS Serverless Application Repository?

AWS Serverless Application Repository is a collection of pre-built, serverless applications that developers can use to quickly deploy serverless solutions like Lambda functions and API Gateway configurations.

Pricing, Billing & Cost Management (continued)

What is AWS Cost and Usage Report?

AWS Cost and Usage Report (CUR) provides detailed information about your AWS usage and costs. It allows you to analyze cost data, optimize usage, and forecast future expenditures.

What is AWS Savings Plans?

AWS Savings Plans are flexible pricing models that provide significant savings on compute usage in exchange for committing to a one- or three-year term. They apply to services like EC2, Lambda, and Fargate.

What is the AWS Well-Architected Framework?

AWS Well-Architected Framework provides best practices for designing and operating reliable, secure, efficient, and cost-effective systems in the cloud. It covers five pillars: operational excellence, security, reliability, performance efficiency, and cost optimization.

What is AWS Pricing Model?

AWS offers several pricing models, including:

Pay-as-you-go: Pay only for the services you use.

Reserved: Commit to using certain services for a one- or three-year term at a discounted rate.

Spot Instances: Bid for unused EC2 capacity at a lower price.

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General Concepts (continued)

What is a VPC Endpoint?

VPC Endpoints allow private connections between your VPC and supported AWS services without requiring access over the internet. There are two types: Gateway Endpoints (for S3 and DynamoDB) and Interface Endpoints (for other AWS services).

What is Amazon Elastic IP (EIP)?

Elastic IP (EIP) is a static IP address designed for dynamic cloud computing. It allows you to remap a public IP address to any instance in your VPC.

What is Amazon Route 53?

Amazon Route 53 is a scalable and highly available Domain Name System (DNS) web service that allows you to route end-user requests to your application. It supports domain registration, DNS routing, and health checks.

What is AWS IAM Access Analyzer?

IAM Access Analyzer helps you identify any resources in your AWS environment that are shared with an external entity (outside your AWS account), providing visibility and helping maintain secure access controls.

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Additional Services

What is AWS Elastic Beanstalk?

AWS Elastic Beanstalk is an easy-to-use service for deploying and managing web applications and services. It automatically handles deployment, capacity provisioning, load balancing, and scaling.

What is AWS Fargate?

AWS Fargate is a serverless compute engine for containers that works with Amazon ECS and EKS. It allows you to run containers without managing the underlying infrastructure.

What is Amazon Lightsail?

Amazon Lightsail is a simplified cloud computing platform designed for smaller businesses, developers, and users who want an easy-to-use, cost-effective option to deploy virtual servers, storage, and networking services.

What is Amazon Elastic Container Registry (ECR)?

Amazon ECR is a fully managed Docker container registry that allows developers to store, manage, and deploy container images for their applications.

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Final Clarifications

What is AWS CloudWatch?

AWS CloudWatch is a monitoring service that provides data and insights on AWS resources and applications. It helps you track performance metrics, monitor logs, and set alarms for automated responses.

What is AWS Trusted Advisor?

AWS Trusted Advisor is a resource optimization tool that helps you follow AWS best practices in areas like cost optimization, performance, security, and fault tolerance. It provides recommendations to improve your AWS infrastructure.

Additional AWS Services & Concepts

AWS CloudFormation

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What is AWS CloudFormation?

AWS CloudFormation is a service that allows you to model and set up your AWS resources so that you can spend less time managing those resources and more time focusing on your applications. You can define your infrastructure as code using JSON or YAML templates.

AWS Lambda

What is AWS Lambda?

AWS Lambda is a serverless compute service that lets you run code in response to events without provisioning or managing servers. Lambda automatically scales your application by running code in response to events such as changes in data, shifts in system state, or user actions.

AWS Elastic Load Balancer (ELB)

What is AWS Elastic Load Balancer (ELB)?

AWS Elastic Load Balancer (ELB) is a fully managed load balancing service that automatically distributes incoming traffic across multiple targets, such as EC2 instances, containers, and IP addresses. ELB supports three types: Classic Load Balancer, Application Load Balancer, and Network Load Balancer.

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Amazon CloudFront

What is Amazon CloudFront?

Amazon CloudFront is a content delivery network (CDN) service that delivers your content with low latency and high transfer speeds. It caches content at edge locations globally, reducing the load on your origin servers and improving user experience.

Identity & Access Management (IAM)

What is an IAM Role in AWS?

An **IAM Role** is an AWS identity with specific permissions. It can be assumed by entities such as users, applications, or services to perform actions on AWS resources. Roles are often used for cross-account access or to grant temporary access to AWS resources.

What is the difference between IAM User and IAM Role?

IAM User is an identity that represents a person or service within your AWS account with specific permissions. An **IAM Role** is not tied to a single user or service but can be assumed by a user or service to perform specific tasks based on policies.

AWS Elastic Beanstalk vs AWS Lambda

What is the difference between AWS Elastic Beanstalk and AWS Lambda?

AWS Elastic Beanstalk is a Platform-as-a-Service (PaaS) that handles deployment and scaling of web applications, while **AWS Lambda** is a serverless compute service where you only pay for the execution of code in response to events. Elastic Beanstalk requires managing application environments, while Lambda handles everything behind the scenes.

Amazon RDS and Databases

What is Amazon RDS?

Amazon RDS (Relational Database Service) is a fully managed relational database service that supports multiple database engines, including MySQL, PostgreSQL, MariaDB, Oracle, and SQL Server. It handles database administration tasks such as patching, backups, and scaling.

What is Amazon Aurora?

Amazon Aurora is a MySQL and PostgreSQL-compatible relational database that is designed for cloud-native applications. It offers improved performance, scalability, and availability compared to traditional relational databases.

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Serverless & Container Services

What is AWS AppSync?

AWS AppSync is a fully managed service that makes it easy to develop GraphQL APIs by handling the complexities of securely connecting to data sources like AWS DynamoDB, Lambda, and more. It simplifies real-time data synchronization and offline access.

What is Amazon ECS?

Amazon ECS (Elastic Container Service) is a fully managed container orchestration service that supports Docker containers. It allows you to easily run and scale containerized applications on AWS.

What is Amazon EKS?

Amazon EKS (Elastic Kubernetes Service) is a managed service that makes it easy to run Kubernetes on AWS without needing to install and operate your own Kubernetes control plane.

Cloud Adoption Framework & Cloud Migration

What is the AWS Cloud Adoption Framework (CAF)?

AWS CAF is a comprehensive framework to help organizations with their cloud adoption journey. It is divided into six perspectives: business, people, governance, platform, security, and operations. These perspectives guide organizations in planning and executing their cloud strategy.

What is the AWS Migration Hub?

AWS Migration Hub provides a central location to track the progress of application migrations across multiple AWS and partner tools. It helps you monitor your cloud migration journey from start to finish.

AWS Support Plans

What are the different AWS Support Plans?

AWS Support Plans are designed to provide varying levels of support:

Basic: Free, offers customer service and access to AWS resources.

Developer: For users experimenting or testing AWS; includes business hours support.

Business: 24/7 access to Cloud Support Engineers and additional technical resources.

Enterprise: Offers a dedicated Technical Account Manager (TAM) and enhanced support.

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Event-Driven Architecture

What is an EventBridge in AWS?

Amazon EventBridge is a serverless event bus service that facilitates event-driven architectures by connecting applications using events from AWS services, integrated SaaS applications, and custom events. It allows for building real-time event-driven systems that can respond to changes.

Other General Concepts

What is a subnet in AWS VPC?

A **Subnet** is a segment of a VPC's IP address range where you can place resources such as EC2 instances. Subnets are typically categorized into public or private subnets, depending on whether they need access to the internet.

What is a VPN Gateway in AWS?

A **VPN Gateway** is a virtual private gateway that allows you to establish an encrypted connection between your on-premises network and your AWS VPC over the internet.

General Cloud Computing Concepts

What is a hybrid cloud?

A **Hybrid Cloud** is an IT environment that combines on-premises infrastructure with cloud services, enabling data and applications to be shared between them. It allows for greater flexibility and deployment options.

What is multi-cloud?

Multi-cloud refers to using services from multiple cloud providers, such as AWS, Azure, and Google Cloud, to avoid vendor lock-in, increase redundancy, or leverage the best services from each provider.

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Monitoring and Management

What is AWS CloudTrail?

AWS CloudTrail is a service that records API calls made on your account. It logs events from different AWS services, allowing you to monitor and track user activity and resource changes for security and auditing purposes.

What is Amazon Inspector?

Amazon Inspector is an automated security assessment service that helps identify vulnerabilities and deviations from best practices in your AWS environment.

Specialized AWS Services and Concepts

Amazon S3 Advanced Features

What is S3 Versioning?

S3 Versioning allows you to preserve, retrieve, and restore every version of every object stored in an S3 bucket. This is useful for data recovery and protects against accidental deletion or overwriting.

What is S3 Cross-Region Replication (CRR)?

S3 Cross-Region Replication allows automatic, asynchronous copying of objects between S3 buckets in different AWS regions. It's often used for disaster recovery, compliance, and latency-based access optimization.

What is S3 Transfer Acceleration?

S3 Transfer Acceleration speeds up file uploads to S3 by utilizing Amazon CloudFront's globally distributed edge locations. It's ideal for transferring large objects over long distances.

AWS Security Services

What is AWS Secrets Manager?

AWS Secrets Manager helps you protect access to applications, services, and IT resources by securely storing and managing secrets such as database credentials, API keys, and other sensitive data.

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What is AWS KMS?

AWS Key Management Service (KMS) is a managed service that helps you create and control encryption keys used to encrypt your data. KMS integrates with many AWS services and is critical for enforcing security best practices.

What is AWS Shield?

AWS Shield is a managed Distributed Denial of Service (DDoS) protection service that safeguards applications running on AWS from DDoS attacks.

Networking and Connectivity

What is AWS Direct Connect?

AWS Direct Connect provides a dedicated network connection from your premises to AWS, reducing latency and increasing bandwidth for large-scale data transfers between your on-premises data center and AWS.

What is AWS Transit Gateway?

AWS Transit Gateway enables you to connect multiple VPCs and on-premises networks through a single gateway. This simplifies the network architecture and reduces the number of peering connections you need to manage.

AWS Analytics Services

What is AWS Glue?

AWS Glue is a fully managed ETL (Extract, Transform, Load) service that makes it easy for you to prepare and transform data for analytics, machine learning, and application development.

What is Amazon Redshift Spectrum?

Amazon Redshift Spectrum allows you to run SQL queries directly against data stored in Amazon S3, extending your Amazon Redshift data warehouse queries to exabytes of data.

What is AWS QuickSight?

AWS QuickSight is a fast, cloud-powered business analytics service that lets you visualize and analyze your data, and share insights across your organization.

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AWS DevOps and Automation

What is AWS CodePipeline?

AWS CodePipeline is a continuous integration and continuous delivery (CI/CD) service for fast and reliable application and infrastructure updates. It automates the build, test, and deploy phases of your release pipeline.

What is AWS CodeDeploy?

AWS CodeDeploy automates application deployments to a variety of compute services, including EC2 instances, Lambda, and on-premises servers, enabling you to deploy your application faster and more reliably.

What is AWS CodeBuild?

AWS CodeBuild is a fully managed build service that compiles source code, runs tests, and produces software packages ready for deployment. It eliminates the need to manage your own build servers.

Machine Learning & AI

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What is AWS SageMaker?

AWS SageMaker is a fully managed service that provides every developer and data scientist with the ability to build, train, and deploy machine learning models quickly.

What is AWS Lex?

Amazon Lex is a service for building conversational interfaces into applications using voice and text. Lex powers the Amazon Alexa virtual assistant.

What is AWS Polly?

Amazon Polly is a text-to-speech service that turns text into lifelike speech, allowing you to create speech-enabled applications.

What is AWS Rekognition?

Amazon Rekognition is a service that makes it easy to add image and video analysis to your applications, such as detecting objects, scenes, faces, and more.

Cost Management

What is AWS Cost Explorer?

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AWS Cost Explorer enables you to visualize, understand, and manage your AWS costs and usage. You can view usage patterns, forecast costs, and create reports to optimize your spending.

What is AWS Trusted Advisor?

AWS Trusted Advisor is an online resource to help you reduce cost, increase performance, and improve security by providing real-time recommendations based on AWS best practices.

Compliance & Governance

What is AWS Artifact?

AWS Artifact provides on-demand access to AWS compliance documentation and AWS agreements, helping you meet governance, risk, and compliance (GRC) requirements.

What is AWS Config?

AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources. It provides a detailed inventory of AWS resources and their configurations, allowing you to track changes and compliance.

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Amazon WorkSpaces and Desktop Services

What is Amazon WorkSpaces?

Amazon WorkSpaces is a managed Desktop-as-a-Service (DaaS) solution that provides cloud-based virtual desktops to users, allowing them to access their desktops from anywhere.

What is Amazon WorkDocs?

Amazon WorkDocs is a fully managed, secure enterprise storage and collaboration service that allows you to store, share, and access documents.

Terraform

Secrets - used to pass the credentials
count - we can launch multiple instances
for_each - function used for creation multiple resources
Access key & Secret Key
Terraform directory structure

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- Configure the log level by setting the TF_LOG environment variable to TRACE, DEBUG, INFO, WARN or ERROR. The TRACE level is the most verbose.
- Set the log-file path by using the TF_LOG_PATH environment variable.

This **state file** keeps track of resources created by your configuration and maps them to real-world resources.

1. fmt - When I've finished my Terraform configuration, I want to make sure that everything is formatted correctly, so I run the fmt command first. This command **reformats your configuration in the standard style**, so it'll make sure that the spacing and everything else is formatted correctly. If it comes back blank, that means the configuration files within your working directory are already correctly formatted. If it does format a file, it will let you know what file it touched.

2. init - After you use the format command, you'll want to initialize your working directory to prepare it for what you need. The init command looks at your configuration files and determines which providers and modules it needs to pull down from the registry to allow your configuration to work properly.

3. validate - Once you've initialised the directory, it's good to run the validate command before you run plan or apply. Validation will catch syntax errors, version errors, and other

issues. One thing to note here is that you can't run validate before you run the init command. You have to initialise the working directory before you can run the validation.

4. plan - Next, it's always a good idea to do a dry run of your plan to see what it's actually going to do. You can even use one of the subcommands with terraform plan to output your plan to apply it later.

5**. apply** - It is one of the commands you're going to use the most. This is the command that deploys or applies your configuration to a provider.

6. destroy - The destroy command, obviously, will destroy your infrastructure — or, when used with the target flag, individual resources within your infrastructure.

7. output - If you've put together a good output variable file, you can use the output command to make those defined outputs to display certain information. For example, if you're deploying EC2 instances, **you can output tag names, instance names, instance IDs, the IP of the instance**, and so on. You can gather some really good information that makes it simple to look up later. And if you're working as a team, people coming behind you can use the output command to figure things out and get up to speed.

8. show - The show command **shows the current state of a saved plan, providing good information about the infrastructure you've deployed**. For example, if you have an EC2 instance or a VM deployed in your configuration, it'll show you the state that it's in — if it's up and ready or if it's being terminated. It also provides useful information like IP addresses.

9. state - Another good way to check your work is to use the state command. If you use state and then the subcommand list, it'll give you a **consolidated list of the resources that are being managed by your configuration**. If you are moving your Terraform instance, such as from a local instance to a remote backup, you would use the state mv command. And just like the show command, there's a state show command that shows a resource in the state. You can also remove instances from a state by using the state rm command.

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10. version - You'll use the version command quite a bit to check your Terraform version, especially if you have any version conflicts. Sometimes providers work only with certain versions of Terraform, so if you're defining those versions within your configuration, you'll need to use the version command here and there.

CMD and **ENTRYPOINT** - execution-wise, the default **arguments** provided by **CMD can get overridden,** whereas those provided by ENTRYPOINT cannot.

CMD is the default argument to container.
CMD commands are **ignored by Daemon when there are parameters stated within the docker run** command.

ENTRYPOINT - instructions in a dockerfile. The ENTRYPOINT instruction is **used to configure the executables that will always run after the container is initiated**.

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ENTRYPOINT instructions are not ignored but instead are appended as command line parameters by treating those as arguments of the command

NOTE - We can override the default entrypoint from the CMD override page in our job or service by **selecting custom command** as the Docker runtime mode.

Build - creates a docker image based on the Dockerfile .
Run - uses the created docker image to run a container.

Created - a container that has been created (e.g. with docker create) but not started.
Restarting - a container that is in the process of being restarted.
Running - a currently running container.
Paused - a container whose processes have been paused.

list containers - **docker container ls**
list images - **docker image ls**
list networks - **docker network ls**
list volumes - **docker volume ls**

Stop all running containers - **docker stop $(docker ps)**

Cleanup containers ≥ v1.13+ - **docker container prune**
Cleanup whole system (network, volumes, etc.) - **docker system prune --all**

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How to Commit Changes to a Docker Image with Examples
1. Step 1: Pull a Docker Image.
2. Step 2: Deploy the Container.
3. Step 3: Modify the Container.
4. Step 4: Commit Changes to Image.

Every-time you run **docker commit** , you will create a new image. On the other hand, **docker build** create the image by referring to a script (Dockerfile). Generally, docker build will only create a new image if it detects changes.

ADD directive can accept a remote URL for its source argument. The COPY directive, on the other hand, can only accept local files. Note that using ADD to fetch remote files and copying is not typically ideal. This is because the file will increase the overall Docker image size.

Kubectl - It is a command line tool used to interact with the Kubernetes cluster. The control plane consists of API server, the etcd metadata store, Scheduler, Controller manager and cloud controller manager.

API Server - The Kube API Server is REST server which exposes Kubernetes APIs to the external world. We can have multiple instances of API server in the control plane for high availability. API server cluster state is kept in etcd.

etcd Store - The complete cluster is stored in etcd (https://coreos.com/etcd/) , which is a consistent, reliable and distributed key value store. The etcd store is an open-source object developed by CoreOS initially. For redundancy 3 to 5 instances are maintained if you lose data in etcd store you lose your cluster also.

Scheduler - Kube-scheduler is responsible for scheduling Pods to worker nodes. Implementation of sophisticated scheduling algorithms takes a lot of information into account such as availability of resources on each node, constraints mentioned by user, type of available nodes, resource limits and quotas, and other factors such as affinity , anti-affinity, tolerations and taints.

Controller Manager - The Kube controller manager is a single process which has multiple controllers to achieve simplicity. These controllers keep a watch on changes and respond accordingly
Node controllers are responsible for notice and respond when a node goes down
Replication controller ensure there is correct number of pods for each replica set or objects for replication controller
Endpoints controller assign for each service an endpoint object which lists the service pods
Service account and token controllers initialize new namespaces with default service account and corresponding API access tokens

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The Data Plane - The data plane comprises nodes in the cluster which run containerized workloads or pods. The data plane and control plane share physical and logical machines. In order to communicate to nodes Kubernetes installs several components on each node, watch, schedule pods etc

Kubelet - The Kubelet is Kubernetes agent and responsible for talking to API servers and running and managing pods on the node. It is responsible for : Download pod secrets from API server, Volume mounts, Pod container running via container runtime interface (CRI), Report status of each Pod and node, Probe container liveness

Kube Proxy - Networking aspects are handled by this component for the node. It operates as local front for services and forwards TCP and UDP packets and discovers IP addresses of services via DNS or environment variables

Container Runtime - Kubernetes runs containers and supports a variety of container runtimes. Docker is supported initially. Kubernetes runs containers via an interface known as CRI which is based on gRPC.

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Call on +91 9355165500